

# Advait Paranjpe

Los Angeles, CA

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## Education

### University of California, Los Angeles

*M.S. in Electrical and Computer Engineering; GPA: 3.7/4.0*

Expected Dec 2026

Los Angeles, CA

- Coursework: CPU/GPU Architecture, Memory Systems, Cache Coherence, Interconnect Design.

### The University of Manchester

*B.Eng. in Electrical and Electronic Engineering; First Class Honours*

Sep 2022 – Jul 2025

Manchester, UK

- Dissertation: 10 GHz custom RF PCB and firmware for 5G systems.

## Experience

### Jaguar Land Rover North America

*Semiconductor Applications Engineering Intern*

Jul 2026 – Sep 2026

Detroit, MI

- Building a Cortex-A76 profiler that helps engineers identify cache, memory, and multicore bottlenecks in automotive SoCs.
- Developed low-level C/C++ Linux tooling for PMU event setup, sampling, workload profiling, and board-level validation.
- Engineered a profiler with under 1% measured runtime overhead across synthetic workloads on a four-core Renesas R-Car V4H.
- Collaborating cross-functionally with platform and semiconductor teams to define event plans, review results, and prioritize validation.

### VAST Lab, UCLA

*Student Researcher – FPGA Architecture & GPU Acceleration*

Jan 2026 – Mar 2026

Los Angeles, CA

- Benchmarked bucketed 128-byte gathers on HBM, reaching about 182 GB/s across roughly 1M lookups into a 4 GB table.
- Swept FPGA cache size and associativity; a 40 MB high-hit-rate design motivated an HBM-backed hot-data cache.

### NESO / National Grid ESO

*Software Engineering Intern*

Jul 2024 – Sep 2025

Leamington Spa, UK

- Refactored a 100k+ LOC Python simulation workflow with vectorized dataflow, cutting runtime by 10x and saving 450+ hours/year.
- Delivered 24x faster pipeline execution using parallel API batching and deterministic processing stages for reproducible model inputs.
- Parallelized 100M+ row workloads through chunked execution, cutting RAM from 32 GB to 16 GB and improving runtime by 11x.

## Projects

### tinyNPU | SystemVerilog

*Systolic Matrix-Multiply Accelerator with ASIC Flow*

May 2026 – Jun 2026

[github.com/AdvaitParanjpe/tinyNPU](https://github.com/AdvaitParanjpe/tinyNPU)

- Designed and verified an INT8 systolic matrix-multiply accelerator, carrying the design from RTL through an ASIC layout flow.
- Implemented a double-buffered streaming architecture with independent load, compute, and output FSMs, sustaining 64 cycles/tile.
- Closed post-route timing at 100 MHz, improving WNS from  $-5.83$  ns to  $+0.89$  ns through architectural redesign.

### Sparrow-V/Cluster | SystemVerilog, C, Python

*RISC-V Processor & Multicore SoC*

Mar 2026 – Jun 2026

[github.com/AdvaitParanjpe/sparrow-v](https://github.com/AdvaitParanjpe/sparrow-v)

- Built a programmable RISC-V edge-computing platform spanning a vector processor and four-core cache-coherent SoC.
- Added four-lane INT8 dense/sparse instructions, cutting 16x4 workload execution by 15.3x and metadata storage by 40.6%.
- Integrated private L1 caches, shared SRAM, snoop MSI, and LR/SC; verified 168 randomized operations plus 10 multicore workloads.

### AXI4-Stream Router IP | SystemVerilog, UVM

*RTL Design & UVM Verification*

Jun 2026 – Jul 2026

[github.com/AdvaitParanjpe/axi-router](https://github.com/AdvaitParanjpe/axi-router)

- Designed a 2x4 AXI4-Stream router with 64-beat buffers, round-robin arbitration, packet locking, backpressure, and drop counters.
- Built conventional and UVM environments with agents, reference models, scoreboards, protocol checks, and randomized backpressure.
- Passed 32 seeded runs plus directed tests covering routing, contention, backpressure, resets, counter wrap, and head-of-line blocking.

## Skills

**Languages:** SystemVerilog, Verilog, VHDL, Tcl, Python, C, C++, Bash, SQL, MATLAB.

**Tools:** Verilator, Vivado, Yosys, OpenLane, GTKWave, Git, Linux, Docker.

**Interests:** Classical and Electric Guitar, Running, Weightlifting, Boulderling.